

Task 4 — Final Summary

Customer Churn Analysis

Key Findings

The analysis covers **7,043 customers**, including **1,869 churned customers** and **5,174 retained customers**. The overall churn rate is **26.53%**, showing that customer retention is an important area for the business.

The analysis indicates that churn varies across different customer characteristics. **Month-to-month contract customers**, **customers with shorter tenure**, and **electronic check users** appear to be important groups for further attention. Monthly charges also differ between churned and retained customers.

Statistical Analysis

Two statistical tests were performed to support the findings:

- **Chi-Square Test:** Used to examine whether there is a relationship between **contract type and churn**. The test result should be evaluated using the p-value at the **5% significance level**.
- **Independent Samples T-Test:** Used to compare **average monthly charges between churned and retained customers**. The p-value determines whether the observed difference is statistically significant.

These tests provide statistical support for identifying whether the observed patterns are meaningful rather than simply due to differences in the sample.

Important Customer Segments

The rule-based segmentation identifies several groups that can be considered for targeted retention strategies:

- **New Customers:** Customers with six months or less tenure. These customers may benefit from stronger onboarding and early engagement.
- **High-Risk Customers:** Customers identified as having churned and more than six months of tenure. Their characteristics can help identify patterns associated with customer loss.
- **High-Value Customers:** Customers with monthly charges of **\$80 or more**. Retaining these customers can help protect higher recurring revenue.
- **Loyal Customers:** Customers who do not fall into the above categories and have remained with the company.

Revenue at Risk

The analysis calculates **monthly revenue at risk** based on the monthly charges of churned customers. This metric is important because it represents the recurring revenue associated with customers who have already left.

Recommendations

1. **Focus on month-to-month customers** and explore ways to encourage longer-term contracts.
2. **Strengthen new-customer onboarding** and follow-up during the early stages of the customer relationship.
3. **Monitor high-risk customers** and identify common characteristics that can support early intervention.
4. **Review electronic-check customers** to understand whether payment-related factors may be contributing to churn.
5. **Protect high-value customers** through personalized support and retention offers.
6. **Use the Power BI dashboard regularly** to monitor churn patterns, customer segments, and revenue at risk.

Conclusion

Overall, the analysis highlights several customer groups that require closer attention, particularly **month-to-month, newer, high-risk, and high-value customers**. Combining the statistical results with dashboard insights can help the business make more informed retention decisions and reduce potential revenue loss.